

MIAE 221: Materials Science for Engineers

Part 3: Crystal Structures, Unit Cells & Crystallography Master Guide

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering (MIAE)
Lectures 4 & 5 · Prof. Mamoun Medraj

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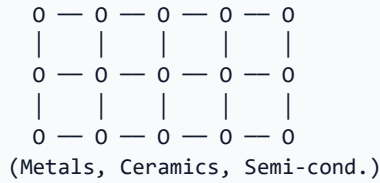
1. Fundamental Concepts of Crystalline Architecture

In solid-state physics and materials engineering, how atoms pack together in three-dimensional space dictates mechanical ductility, electrical conductivity, magnetic permeability, and deformation slip planes.

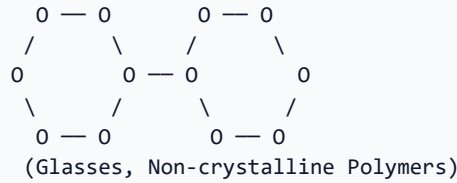
Crystalline vs. Amorphous Solids

Solids are divided into two fundamental structural classes based on the regularity of their atomic arrangements:

[CRYSTALLINE SOLID]
(Periodic, Long-Range Order)



[AMORPHOUS SOLID]
(Non-Periodic, Short-Range Order Only)



1. Crystalline Materials:

- Atoms position themselves in a **repeating, periodic 3D array** spanning large atomic distances (long-range order).
- Upon solidification, thermodynamic energy is minimized when atoms nestle into predictable geometric equilibrium sites.
- Examples:* All engineering metals (Fe, Al, Cu, Ti), most technical ceramics (Al_2O_3 , SiC), and semiconductors (Si, GaAs).

2. Amorphous (Non-Crystalline / Vitreous) Materials:

- Atoms lack systemic long-range spatial periodicity; only localized chemical bond distances (short-range order) exist.
- Occurs when rapid liquid cooling ("quenching") freezes atoms in place before they have time to arrange into equilibrium crystal lattices.
- Examples:* Silica window glass, amorphous polymers (polystyrene, PMMA), and amorphous metallic glass alloys.

Lattices, Unit Cells & Motifs

To mathematically describe crystals, crystallographers separate geometry from atomic identity:

- Lattice:** An infinite 3-dimensional array of mathematical points in space, wherein every point has identical surroundings.
- Basis / Motif:** An atom, ion, or cluster of atoms attached identically to every single lattice point.

$$\text{Crystal Structure} = \text{Lattice} + \text{Basis}$$

- Unit Cell:** The **smallest structural repeating unit** that completely defines the symmetry and crystal architecture. Stacking identical unit cells edge-to-edge in 3 dimensions recreates the entire macroscopic crystal lattice.

The Atomic Contribution Counting Law

Because unit cells in a crystal share corners, edges, and faces with neighboring unit cells, an atom sitting on a boundary does **not** belong entirely to one single cell.

To determine the net effective number of atoms (N) inside one unit cell:

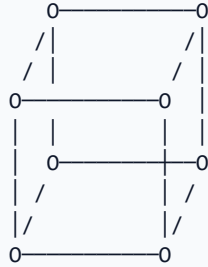
$$N = N_{\text{interior}} + \frac{N_{\text{face}}}{2} + \frac{N_{\text{edge}}}{4} + \frac{N_{\text{corner}}}{8}$$

Atom Position	Shared Between How Many Cells?	Fractional Contribution to One Cell
Interior / Center (N_i)	1 (Completely inside)	1
Face Center (N_f)	2 adjacent unit cells	$\frac{1}{2}$
Edge Center (N_e)	4 adjacent unit cells	$\frac{1}{4}$
Corner (N_c)	8 adjacent unit cells	$\frac{1}{8}$

2. The Major Metallic Crystal Structures

Over 90 of all elemental metals crystallize into one of three densely packed geometries: **BCC**, **FCC**, or **HCP**. A fourth theoretical structure, **Simple Cubic (SC)**, provides the geometric baseline.

A. Simple Cubic (SC)



Cube Edge: a
 Atomic Radius: R
 Atoms touch along cube edges:
 $a = 2R$

1. Lattice Geometry & Atom Contact:

- Atoms reside **only** at the 8 cube corners.
- Atoms touch each other directly along the **cube edges**:

$$a = 2R$$

2. Coordination Number (CN):

- Each atom is in direct contact with **6** nearest neighbors ($\pm x, \pm y, \pm z$).

3. Number of Atoms Per Unit Cell (N):

$$N = 8 \text{ corners} \times \frac{1}{8} = 1 \text{ atom/cell}$$

4. Atomic Packing Factor (APF):

- Definition:** The fraction of solid sphere volume occupied inside the unit cell:

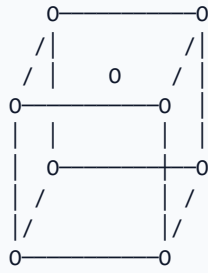
$$\text{APF} = \frac{V_{\text{atoms}}}{V_{\text{unit cell}}} = \frac{N \times \left(\frac{4}{3}\pi R^3\right)}{a^3}$$

- Substituting $a = 2R$:

$$\text{APF}_{\text{SC}} = \frac{1 \times \frac{4}{3}\pi R^3}{(2R)^3} = \frac{\frac{4}{3}\pi R^3}{8R^3} = \frac{\pi}{6} \approx \mathbf{0.52}$$

- Assessment:** Only 52 of the space is filled. Because of its extremely loose, unstable packing, only one elemental metal adopts Simple Cubic: **Polonium (Po)**.

B. Body-Centered Cubic (BCC)



Corner atoms + 1 Center atom
Close-packed direction: Body Diagonal
Body Diagonal = $\sqrt{3} \cdot a = 4R$

$$a = 4R / \sqrt{3}$$

1. Lattice Geometry & Atom Contact:

- Atoms are located at the 8 corners plus **1 full atom in the center of the cube**.
- Corner atoms do **not** touch along the cube edge ($a > 2R$).
- Atoms touch continuously across the **cube body diagonal**:

$$\text{Body Diagonal} = \sqrt{a^2 + a^2 + a^2} = a\sqrt{3} = 4R \implies a = \frac{4R}{\sqrt{3}}$$

2. Coordination Number (CN):

- The center atom directly touches all 8 corner atoms $\implies \text{CN} = 8$.

3. Number of Atoms Per Unit Cell (N):

$$N = 1 (\text{center}) + 8 \text{ corners} \times \frac{1}{8} = 2 \text{ atoms/cell}$$

4. Unit Cell Volume (V_c):

$$V_c = a^3 = \left(\frac{4R}{\sqrt{3}} \right)^3 = \frac{64R^3}{3\sqrt{3}}$$

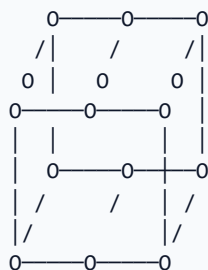
5. Atomic Packing Factor (APF):

$$\text{APF}_{\text{BCC}} = \frac{2 \times \left(\frac{4}{3}\pi R^3 \right)}{\frac{64R^3}{3\sqrt{3}}} = \frac{\frac{8}{3}\pi R^3}{\frac{64}{3\sqrt{3}}R^3} = \frac{\pi\sqrt{3}}{8} \approx 0.68$$

6. Engineering Examples:

- α -Iron (ferrite at room temp), Chromium (Cr), Tungsten (W), Molybdenum (Mo), Tantalum (Ta), Vanadium (V).

C. Face-Centered Cubic (FCC)



Corner atoms + 6 Face-Center atoms
Close-packed direction: Face Diagonal
Face Diagonal = $\sqrt{2} \cdot a = 4R$

$$a = 2\sqrt{2} \cdot R = 4R / \sqrt{2}$$

1. Lattice Geometry & Atom Contact:

- Atoms reside at the 8 corners plus **in the center of all 6 cube faces**.
- Atoms touch continuously along the **face diagonal**:

$$\text{Face Diagonal} = \sqrt{a^2 + a^2} = a\sqrt{2} = 4R \implies a = 2\sqrt{2}R = \frac{4R}{\sqrt{2}}$$

2. Coordination Number (CN):

- Each atom touches 4 neighbors in its own plane, 4 in the plane above, and 4 in the plane below $\implies \text{CN} = 12$.

3. Number of Atoms Per Unit Cell (N):

$$N = 8 \text{ corners} \times \frac{1}{8} + 6 \text{ faces} \times \frac{1}{2} = 1 + 3 = 4 \text{ atoms/cell}$$

4. Unit Cell Volume (V_c):

$$V_c = a^3 = (2\sqrt{2}R)^3 = 16\sqrt{2}R^3$$

5. Atomic Packing Factor (APF):

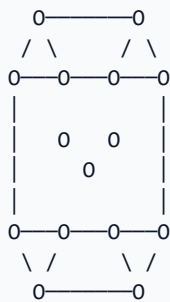
$$\text{APF}_{\text{FCC}} = \frac{4 \times \left(\frac{4}{3}\pi R^3\right)}{16\sqrt{2}R^3} = \frac{\frac{16}{3}\pi R^3}{16\sqrt{2}R^3} = \frac{\pi}{3\sqrt{2}} \approx 0.74$$

- *Significance*: 0.74 represents the **mathematical upper limit** for packing equal, hard spheres in 3D Euclidean space (Kepler conjecture). FCC is a **close-packed** structure.

6. Engineering Examples:

- Aluminum (Al), Copper (Cu), Gold (Au), Silver (Ag), Nickel (Ni), Platinum (Pt), Lead (Pb), γ -Iron (austenite at $T > 912^\circ\text{C}$).

D. Hexagonal Close-Packed (HCP)



Top Basal Plane: 6 corner + 1 center

Mid-Plane: 3 interior atoms in triangle

Bottom Basal Plane: 6 corner + 1 center

1. Lattice Geometry:

- Two parallel hexagonal basal planes separated by height c .
- Basal hexagon edge length is $a = 2R$.
- In an ideal HCP structure, the axial ratio is:

$$\frac{c}{a} = \sqrt{\frac{8}{3}} \approx 1.633$$

2. Coordination Number (CN):

- **CN = 12** (Identical to FCC; 6 neighbors in-plane, 3 above, 3 below).

3. **Number of Atoms Per Unit Cell (*N*):**

- The complete hexagonal prism contains:
 - 12 corners $\times \frac{1}{6}$ (each shared by 6 hexagonal prisms) = 2
 - 2 basal face centers $\times \frac{1}{2} = 1$
 - 3 interior atoms entirely within the cell = 3

$$N_{\text{total}} = 2 + 1 + 3 = \mathbf{6 \text{ atoms/cell}}$$

4. **Unit Cell Volume & APF:**

- The volume of the hexagonal prism is $V_c = \text{Base Area} \times c = \left(6 \times \frac{\sqrt{3}}{4}a^2\right)c = \frac{3\sqrt{3}}{2}a^2c$.
- Substituting $a = 2R$ and $c = 1.633a$:

$$\text{APF}_{\text{HCP}} = \mathbf{0.74} \quad (\text{Identical to FCC!})$$

5. **Engineering Examples:**

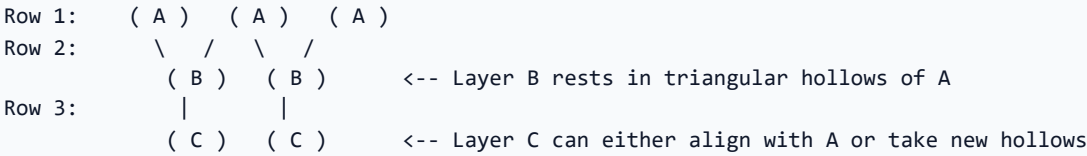
- Titanium (α -Ti), Magnesium (Mg), Zinc (Zn), Cobalt (Co), Zirconium (Zr), Beryllium (Be).

3. **Close-Packed Planes & Stacking Sequences**

Both FCC and HCP achieve the maximum theoretical packing density of $\text{APF} = 0.74$ and $\text{CN} = 12$. Why, then, are they distinct crystal structures with radically different mechanical ductilities?

The answer lies in **layer stacking order**.

The Close-Packed Layer of Spheres



- When you place a 2D sheet of spheres together as tightly as possible, each sphere touches 6 neighbors, forming a triangular array of "valleys" or hollows.
- Let the first layer be **Layer A**.
- The second layer (**Layer B**) rests naturally in the triangular indentations of Layer A.

HCP vs. FCC: The Third Layer Decides

When placing the **third layer of spheres**, there are two distinct choices:

Crystal System	Stacking Sequence	Third Layer Placement	Ductility & Slip Systems
HCP	ABABAB...	Placed directly over the spheres of Layer A .	Limited Ductility: Slip restricted primarily to basal plane 0001. Brittle/hard at room temp.
FCC	ABCABC...	Placed over the <i>alternate unoccupied hollows</i> (Layer C). Does not align with A until the 4th layer.	Exceptional Ductility: 12 identical $111\langle 110 \rangle$ close-packed slip systems allow effortless dislocation glide.

Key Takeaway:

- **HCP** is stacked along the $[0001]$ c -axis direction in an **ABAB...** sequence.
- **FCC** close-packed planes are the 111 family, stacked along the cube body diagonal $[111]$ in an **ABCABC...** sequence.

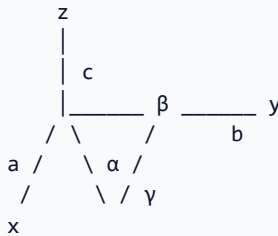
4. The 7 Crystal Systems & 14 Bravais Lattices

Every possible repeating crystalline network can be categorized by the geometry of its unit cell.

Lattice Parameters: Edge Lengths & Interaxial Angles

A unit cell is geometrically defined by 6 independent lattice parameters:

1. Three axial edge lengths: **a, b, c**
2. Three interaxial angles:
 - α : Angle between b and c
 - β : Angle between a and c
 - γ : Angle between a and b



The Master 7 Crystal Systems Matrix

THE 7 UNIQUE CRYSTAL SYSTEMS		
System	Axial Edge Lengths	Interaxial Angles
1. Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$
2. Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$
3. Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$
4. Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^\circ, \gamma = 120^\circ$
5. Rhombohedral	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$
6. Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90^\circ \neq \beta$
7. Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$

- **14 Bravais Lattices:** When lattice points are placed at corners (Primitive, P), centers (Body-centered, I), faces (Face-centered, F), or end bases (Base-centered, C), only **14 unique non-redundant spatial lattices** are mathematically possible across these 7 systems.

5. Theoretical Density Computations (ρ)

A crystal's macroscopic mass density can be calculated directly from its microscopic unit cell geometry.

The Governing Formula & Dimensional Analysis

$$\rho = \frac{n \cdot A}{V_c \cdot N_A}$$

Variable	Description	Standard Engineering Units
ρ	Theoretical density of the solid metal	g/cm ³
n	Number of atoms associated with each unit cell	atoms/cell (1 for SC, 2 for BCC, 4 for FCC, 6 for HCP)
A	Atomic weight of the element	g/mol (or amu)
V_c	Unit cell volume (a^3 for cubic systems)	cm ³ /cell (Convert nm $\rightarrow$ cm: 1 nm = 10 ⁻⁷ cm)
N_A	Avogadro's Number	6.022 $\times$ 10 ²³ atoms/mol

Step-by-Step Numerical Example (FCC Copper)

Problem: Copper (Cu) has an atomic radius $R = 0.128 \text{ nm}$ ($1.28 \times 10^{-8} \text{ cm}$) and an atomic weight $A = 63.55 \text{ g/mol}$. It crystallizes into an FCC lattice. Compute its theoretical density.

Step 1: Identify Parameters for FCC:

- $n = 4 \text{ atoms/cell}$
- $a = 2\sqrt{2}R = 2\sqrt{2}(0.128 \times 10^{-7} \text{ cm}) = 3.62 \times 10^{-8} \text{ cm}$

Step 2: Calculate Unit Cell Volume (V_c):

$$V_c = a^3 = (3.62 \times 10^{-8} \text{ cm})^3 = 4.74 \times 10^{-23} \text{ cm}^3$$

Step 3: Solve for Density (ρ):

$$\rho = \frac{n \cdot A}{V_c \cdot N_A} = \frac{(4)(63.55 \text{ g/mol})}{(4.74 \times 10^{-23} \text{ cm}^3)(6.022 \times 10^{23} \text{ atoms/mol})} = \frac{254.2}{28.54} = 8.91 \text{ g/cm}^3$$

(Experimental measured density of pure copper is 8.94 g/cm³, matching theoretical calculation within < 0.4 error!)

6. Crystallographic Points, Directions & Planes (Miller Indices)

Plastic deformation (dislocation slip), magnetic alignment, and mechanical cleavage occur along specific crystal directions and atomic planes.

Point Coordinates

Point positions within a unit cell are denoted by dimensionless fractional multiples of the cell edge dimensions:

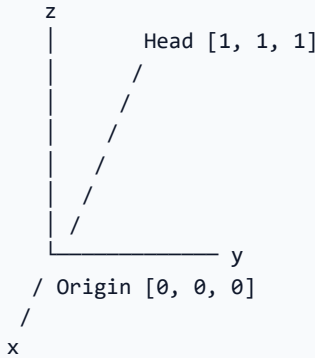
$$\mathbf{q} \quad \mathbf{r} \quad \mathbf{s}$$

Where $x = q \cdot a$, $y = r \cdot b$, and $z = s \cdot c$.

- The origin is $(0, 0, 0)$.
- The center of a BCC unit cell is $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$.
- The center of the bottom face in FCC is $(\frac{1}{2}, \frac{1}{2}, 0)$.

Crystallographic Directions: $[uvw]$ & Families $\langle uvw \rangle$

A direction is a vector joining two points in the lattice, specified inside **square brackets** $[uvw]$.



The 4-Step Vector Algorithm:

1. **Define Coordinates:** Position vector tail at the origin $(0, 0, 0)$ and locate the tip head (x_2, y_2, z_2) .
 2. **Vector Subtraction:** $\Delta x = x_2 - x_1$, $\Delta y = y_2 - y_1$, $\Delta z = z_2 - z_1$ as multiples of a, b, c .
 3. **Clear Fractions:** Multiply or divide by a common factor to reduce numbers to the **smallest integers**.
 4. **Enclose:** Write in square brackets $[u, v, w]$. If an index is negative, represent it with an overbar (e.g., $[\bar{1}, 1, 0]$).
- **Families of Directions $\langle uvw \rangle$:** In a cubic system, because of symmetry, $[100]$, $[010]$, $[001]$, $[\bar{1}00]$, $[0\bar{1}0]$, $[00\bar{1}]$ are physically identical. They are grouped into the family:

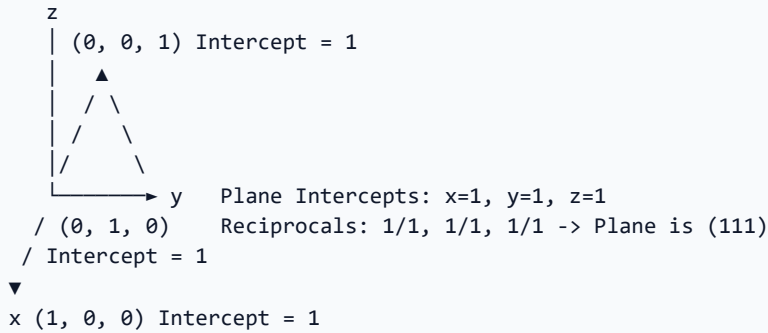
$$\langle 100 \rangle$$

Isotropy vs. Anisotropy in Single Crystals

- **Anisotropic:** Material properties (elastic modulus E , thermal expansion, refractive index) vary depending on the crystallographic direction measured:
 - For single-crystal BCC Iron: $E_{[111]} = 272.7 \text{ GPa}$, but $E_{[100]} = 125.0 \text{ GPa}$!
- **Isotropic:** Properties are independent of measurement direction.
 - While individual crystal grains are anisotropic, **polycrystalline metals** composed of millions of randomly oriented microscopic grains average out to exhibit **macroscopic isotropy**.

Crystallographic Planes: (hkl) & Families hkl

Atomic planes are designated by Miller indices enclosed in **parentheses** (hkl) .



The 4-Step Plane Algorithm:

- Origin Verification:** If the plane passes through the chosen origin $(0, 0, 0)$, **you MUST shift the origin** to an adjacent corner of the unit cell!
- Determine Intercepts:** Identify where the plane intersects the x, y, z axes in terms of lattice parameters a, b, c .
 - If a plane is parallel to an axis, its intercept is ∞ .
- Take Reciprocals:** Invert the intercepts $(\frac{1}{x_{\text{int}}}, \frac{1}{y_{\text{int}}}, \frac{1}{z_{\text{int}}})$.
 - Note: $\frac{1}{\infty} = 0$.
- Clear Fractions & Enclose:** Multiply by a common factor to obtain the smallest integers, and write inside parentheses:

$$(h, k, l)$$

- Negative intercepts receive an overbar (e.g., $(1, \bar{1}, 0)$).

- Families of Planes hkl :** Planes with identical atomic packing density and spacing are grouped into curly braces:

$$100 = (100), (010), (001), (\bar{1}00), (0\bar{1}0), (00\bar{1})$$

7. Comprehensive Comparative Synthesis Matrix & Exam Traps

Master Crystallographic Reference Table

Property	Simple Cubic (SC)	Body-Centered Cubic (BCC)	Face-Centered Cubic (FCC)	Hexagonal Close-Packed (HCP)
Unit Cell Geometry	Cubic	Cubic	Cubic	Hexagonal Prism ($c/a = 1.633$)
Lattice Parameter (a)	$a = 2R$	$a = \frac{4R}{\sqrt{3}}$	$a = 2\sqrt{2}R$	$a = 2R$ ($c = 1.633a$)
Coordination Number (CN)	6	8	12	12
Atoms Per Unit Cell (N)	1	2	4	6
Atomic Packing Factor (APF)	0.52	0.68	0.74 (Max)	0.74 (Max)
Close-Packed Direction	$\langle 100 \rangle$	$\langle 111 \rangle$ (Body diag.)	$\langle 110 \rangle$ (Face diag.)	$\langle 11\bar{2}0 \rangle$ (Basal a-axes)
Close-Packed Plane	None	None	111	0001 (Basal plane)
Stacking Sequence	Single layer repeat	Interpenetrating grids	ABCABC...	ABABAB...

Top 5 High-Yield Exam Traps

[!CAUTION] **Trap 1: Forgetting to Shift Origin for Planes:** If a plane passes directly through the origin (0, 0, 0), you **cannot** write $\frac{1}{0} = \infty$. You must immediately shift the origin by +1 or −1 along an axis to a neighboring unit cell corner before taking intercepts!

[!WARNING] **Trap 2: Mixing Brackets vs. Parentheses:**

- Point coordinate: q, r, s (no brackets or commas)
- Specific direction: $[uvw]$ (square brackets)
- Family of directions: $\langle uvw \rangle$ (angle brackets)
- Specific plane: (hkl) (parentheses)
- Family of planes: hkl (curly braces) Confusing brackets on a midterm results in automatic loss of marks!

[!TIP] **Trap 3: Unit Cell Density Conversion Units:** When using $\rho = \frac{nA}{V_c N_A}$, always convert atomic radius R from nanometers (nm) to centimeters (cm) by multiplying by 10^{-7} cm/nm before cubing a^3 . If you forget, your density will be off by a factor of 10^{21} !

[!NOTE] **Trap 4: HCP Atom Count vs. FCC Atom Count:** Students frequently assume HCP and FCC have the same number of atoms per unit cell because they have the same APF = 0.74. **False!** FCC contains **4 atoms/cell**, whereas a complete HCP unit cell contains **6 atoms/cell**.

[!IMPORTANT] **Trap 5: Close-Packed Planes in BCC:** BCC does **NOT** possess close-packed planes! Its most densely packed planes are 110 , but they have an atomic packing density of only 0.83 (compared to 0.91 for 111 in FCC). This is why BCC metals generally have higher yield strengths and lower ductility than FCC metals.